

Recovering a Hidden Routing Boundary from Query-Only Access

Method note, v5

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Abstract

Given query access to a deployed model and a set of real inputs, we estimate the *geometry* of a hidden routing rule. Around each anchor we send a probe cloud shaped by the local data geometry, remove within-branch trend with a trimmed local-linear fit, and test the residual distribution for a two-component structure whose components are linearly separable in input space. From the fitted mixture, its *weight* gives the distance to the boundary and its *separating direction* gives the boundary's normal; both are checked for consistency across a scale ladder. Pooling these local estimates across anchors—weighted by evidence rather than thresholded—yields a single hyperplane with a confidence interval and a permutation-calibrated p -value, and where that hyperplane is axis-dominated, the coordinate and threshold the system routes on. We characterise where this is identifiable and where it provably is not.

This note supersedes `pipeline_v4` and the Stage 0/1 description in `probe_policy_spec`. The change is structural rather than additive: v4 described a detector with recovery appended. This document is organised around the object being estimated—the boundary—and treats every earlier stage as part of the argument that the estimate is trustworthy.

Provenance tags used throughout.

[VALIDATED]	measured against ground truth
[SPECIFIED]	designed in detail, not implemented
[NEW]	introduced in this note
[OPEN]	acknowledged gap
[LIMIT]	structural impossibility, not expected to yield to engineering

PART I

What the method claims

1 The target sentence

A deployed model may contain a hidden switch: one rule on some inputs, a different rule on others, with nothing in the interface saying so. We have query access only—no weights, no gradients, no routing metadata, no labels.

The claim we want to end up making is not “*something looks suspicious here.*” It is:

The system routes on coordinate i at threshold $\hat{t} = 4.98 \pm 0.07$, recovered from N anchors under query-only access, $p < 0.05$ against a permutation null.

That is an *inverse problem*, not a test. The question is not “does hidden routing happen”—several published methods answer that. The question is “*given only the system’s outputs, what can I infer about the mechanism that produced them.*” Everything in this note exists to make that sentence defensible.

2 The idea, in words

Pick a real input x —call it an **anchor**. Send a small cloud of nearby queries and look at the answers.

If the anchor sits well inside one routing region, every probe gets the same rule and the answers vary smoothly. If the anchor sits near a boundary, some probes land on the far side and get a *different* rule. Those answers are shifted by a constant—the penalty Δ —so the response distribution splits into two clumps rather than one spread.

Three quantities fall out of that split, and each says something different.

- **How many probes crossed** tells you **how far away the boundary is**. If 5% of probes crossed, the boundary is far relative to the probe radius; if 40% crossed, it is close. The Gaussian geometry converts one into the other exactly.
- **Where the crossers sit** tells you **which way the boundary faces**. The mixture model labels each probe as probably-branch-1 or probably-branch-2. Plot those labels back in input space and fit the separating plane; its perpendicular is the boundary’s direction.
- **Agreement across anchors** tells you **whether they are all seeing the same boundary**, and pooling many noisy local planes gives one global plane with error bars.

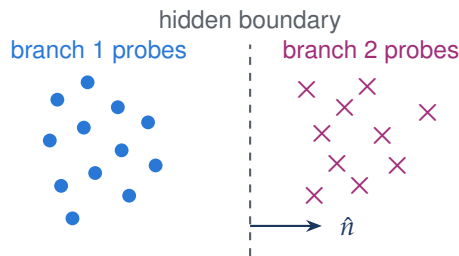


Figure 1: One anchor. The responses say, probabilistically, which side each probe fell on. The plane separating those two groups *in input space* has the boundary’s orientation; its perpendicular is $\hat{n}$.

One anchor gives a noisy local estimate. Repeat around many anchors and pool.

3 A worked example, end to end

Suppose the true hidden rule is $g(x) = \mathbf{1}[x_3 \geq 5]$. You do not know this.

You pick an anchor with $x_3 = 4.7$ and probe with per-coordinate scale $\sigma = 0.5$.

1. The mixture. The two-component fit returns $\hat{\pi} = 0.27$: about 27% of probes landed on the other branch.

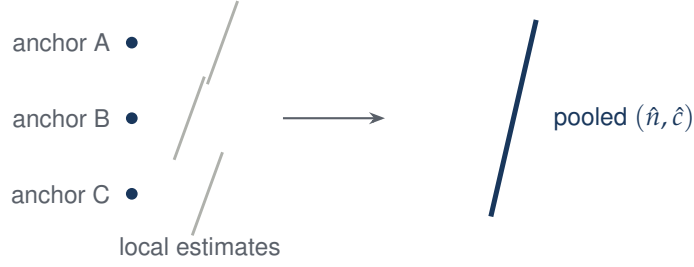


Figure 2: Many anchors. Local planes that are individually noisy pool into one global plane. If the pooled normal is dominated by a single coordinate, the rule is axis-aligned and can be stated as a feature and a threshold.

2. Distance. The signed displacement along any unit direction is $N(0, \sigma^2)$, so the crossing fraction is $\Phi(-d/\sigma)$. Invert it:

$$\hat{d} = -\sigma \Phi^{-1}(\hat{\pi}) = -0.5 \cdot \Phi^{-1}(0.27) = -0.5 \cdot (-0.613) = 0.31.$$

The true distance is $5 - 4.7 = 0.30$. *You have just measured how far away an unobservable boundary is, from nothing but the proportion of a mixture.*

3. Direction. The probes assigned to the second component are mostly those where x_3 increased. The classifier separating the components returns $\hat{n} \approx (0, 0, 1, 0, \dots)$. Note the distance was *unsigned*—the crossing law gives magnitude only, and the direction comes from $\hat{n}$. You need both.

4. Offset. Anchor at $x_3 = 4.70$, 0.31 away in the $+x_3$ direction: $\hat{c} = 4.70 + 0.31 = 5.01$.

5. Pool. Do this at 100 anchors and combine. You might land at $\hat{n} = (0.01, -0.02, 0.999, 0.01, \dots)$ and $\hat{c} = 4.98 \pm 0.07$ —and the claim in §1 is now supported by evidence.

4 Why the earlier stages exist

Steps 1–5 above are the whole scientific content, and every one of them can be wrong for a reason that has nothing to do with routing. The rest of the pipeline is the list of those reasons and what stops each.

So the stages are not scaffolding around a detector. They are the reasons a reader should believe $(\hat{n}, \hat{c})$.

PART II

The machinery

5 Setup and standing assumptions

$$y = f_{g(x)}(x) + \varepsilon, \quad \varepsilon \sim N(0, \tau^2), \quad g: \mathcal{X} \rightarrow \{1, 2\} \text{ hidden}, \quad g(x) = \mathbf{1}[w^\top x > b].$$

Crossing swaps branch and shifts the output by an additive Δ . Routing is **top-1 and hard**—the entering branch arrives at full weight, so the output *steps*. Top-2 softmax routing instead produces a *kink*: the output stays continuous and only the derivative jumps. That is out of scope, and §11 shows the distinction is detectable rather than assumed.

What could go wrong	What stops it
The probe ball never reaches the boundary, so there is nothing to see	Stage 0 — bound the radius by the geometry, and report when no valid radius exists
The probes land where no real input occurs, so the “gate” is real in input space but never fires in deployment	Stage 1 — density filter
The probe cloud inherits the <i>data’s</i> clumpiness and reads as two branches under a perfectly smooth model	Stage 1 — a smooth cloud in the tangent frame is unimodal by construction
A smooth trend across the ball masks the jump—or worse, the fitted plane eats it	Stage 2 — trimmed residualisation
Ordinary curvature produces a lopsided residual distribution that a mixture fits better than one Gaussian	Stage 3 — the separability guard, plus the checks in Stage 5
The mixture is real but it is not a <i>boundary</i> —components interleaved in space	Stage 3 — separability again
The local estimates are individually too noisy to trust	Stage 6 — pool, weighting by evidence rather than thresholding

An anchor set A is drawn from the deployment distribution and **fixed before any query to f** . This is not a convenience: it is what makes the adaptive scale selection in Stage 0 free of any selection penalty. If anchors were chosen using earlier responses, the argument collapses.

6 Stage 0 — where is it legitimate to probe?

Status. Specified, not implemented, and deliberately deferred below the critical path—see §15. The recovery results are established first with the frame and radius *supplied*; this stage is what later lets the method choose them for itself.

Reads anchor covariates only; never touches f . Costs zero queries.

The probe radius must satisfy two things that pull against each other: big enough to reach the boundary, small enough that the honest surface is roughly flat over the ball so the plane fitted in Stage 2 is the right shape.

An intrinsic-dimension curve answers both at once. Estimate $\hat{d}$ at a ladder of neighbour ranks using GRIDE—the local density cancels in a *ratio* of neighbour distances, which is why this family survives clumpy data. The curve has three regimes: inflated at small scales because observation noise is full-dimensional; **flat in the middle—the plateau**; drifting at large scales as the neighbourhood bends or swallows a second cluster. The plateau is the window in which the neighbourhood looks like a flat, evenly-sampled patch.

```

1. NEIGHBORS  N <- k_max nearest neighbours of x in A
               if too few, or kNN radius too large:  ABSTAIN A0_isolated

2. CURVE      for each rank n1 (with n2 = 2*n1):
               d_hat[n1], CI[n1] <- GRIDE(N, n1, n2)
               r[n1] <- median distance to the n1-th neighbour

3. PLATEAU    longest contiguous run of rungs whose CIs intersect
               (j - 1/2, j + 1/2) for a common integer j
               W <- width of that run, in RANK-decades

```

```

        if W < W_min:
            if r[k_max] < 3*tau_hat*sqrt(D): ABSTAIN A1n_budget
            else: ABSTAIN A1_no_plateau

4. SANITY      if d_hat >= gamma*D: FALLBACK to ambient probing

5. FRAME      U <- top d_hat principal directions of neighbours within r_hi

```

Three corrections from the July review are folded in. [\[NEW\]](#)

- The plateau rule is **round-consistency**, not common-intersection. The old rule required all confidence intervals in a run to share a point, which is anti-monotone in data quality: at low noise the intervals are tight, they fail to intersect, and a perfectly flat curve returns “no plateau.” It conflated *statistically indistinguishable* with *geometrically flat*.
- Width is in **rank-decades**, not radius-decades. Since $r \sim k^{1/d}$, a radius-decade threshold is unachievable by construction at $d \geq 5$: ranks 5–200 span 1.6 rank-decades, which is 0.8 radius-decades at $d = 2$ but only 0.32 at $d = 5$.
- **A1 splits from A1n** because they have different remedies. A1 is a property of the anchor—no radius exists at which the assumptions hold, and more budget will not help. A1n is a property of the budget—the ladder never escaped the noise skirt, which extends to roughly $3\text{--}4\tau\sqrt{D}$, not $\sim \tau$. Diagnosable without spending a query.

What Stage 0 cannot tell you: whether the gate is reachable. The plateau is a property of the data’s shape; the gate is a property of f . Bounding the radius from above makes an out-of-window gate more likely to be missed—the method buys validity partly by shrinking where it looks. [\[LIMIT\]](#)

7 Stage 1 — generating probes that mean something

```

ladder    = S rungs, geometric, from max(r_lo, r_hi/10) up to r_hi    # S = 3
sigma_s   = r_s / sqrt(d_hat)                                         # radius -> per-coordinate scale

z_i       ~ N(0, sigma_s^2 I_{d_hat})                                  # in tangent coordinates
delta_i   = U z_i                                                      # ambient displacement
keep_i    = DensityFilter(x + delta_i, A)
rho       = |retained| / m
if rho < rho_min: ABSTAIN A3_off_manifold                             # before any query is spent
y_i       = f(x + delta_i)                                             # the only queries spent

```

The scale conversion is not cosmetic. For $z \sim N(0, \sigma^2 I_d)$ the displacement norm concentrates at $\sigma\sqrt{d}$, so placing a rung at radius r requires $\sigma = r/\sqrt{d}$. Getting this wrong reproduces a validated failure mode: under a fixed *total* budget the ball’s radius stops growing with dimension, the crossing rate becomes $\Phi(-d\sqrt{D}/\sigma)$ and collapses, and **detectability**—not merely power—is destroyed. Detectable anchors fell $100 \rightarrow 78 \rightarrow 56$ at $D = 8, 16$. [\[VALIDATED\]](#)

Isotropic in the frame means round within the $\hat{d}$ estimated directions and zero off them. In ambient $\mathbb{R}^D$ that is a flat pancake—highly anisotropic—but isotropic in the coordinates that matter. Roundness matters because the boundary’s orientation is unknown, so a round cloud has no blind spot.

This is also the fix for the failure that killed the tabular experiment. “On-manifold” covers two different operations, and conflating them is fatal:

Sense	Construction	Effect
follows the manifold’s <i>geometry</i>	smooth cloud in the tangent frame	what we want — in-distribution, unimodal by construction
reproduces the data’s <i>density</i>	interpolation / mixup-style re-sampling	inherits the data’s multimodality $\Rightarrow$ false positives

The second sense drove honest false positives to 0.49 in a controlled setting, and up to 1.00 on real building data whose covariates are themselves multimodal (Hartigan’s dip on `YearBuilt`: $p = 0.0000$ —construction booms). A Gaussian cloud in the frame does not resample the data and does not inherit its clustering.

The density filter never sees y . Retention depends on the displacement and the anchor set only, so dropping probes cannot manufacture response-side structure. What it buys is claim scope: reporting $\hat{\pi}$ on *retained* probes is what separates *a gate exists somewhere in input space* from *a gate fires on inputs the model actually sees*. Only the second is an audit claim, and trimming cannot make that distinction.

$S = 3$, **not 2**. Two rungs cannot separate a gate from resonant curvature: the ratio $\hat{\Delta}(r/2)/\hat{\Delta}(r)$ is 0.94 on a resonant surface against 1.01 for a gate. At $r/4$ the resonant ratio collapses to 0.04. Three rungs make the Stage-5 checks decisive; the cost is $+m$ queries per anchor.

8 Stage 2 — keeping the fit off the minority branch

Least trimmed squares with $h = 0.75m$, fitted in tangent coordinates:

```
1. fit y ~ a + b'z on all retained probes
2. drop the worst-fitting 25% by |residual|
3. refit on the remaining 75%
4. residualise ALL points against the refit plane
```

The mechanism is not variance reduction—it is **keeping the fitted plane on one branch**. Plain OLS finds a compromise plane tilted between the two branches and absorbs

$$\Delta \left(1 - 2\phi(0)\sqrt{2/\pi} \right) \approx 0.36 \Delta,$$

about **64% of the gap** at balanced mixing. Trimming discards points that are predominantly the minority branch, so the plane commits to the majority and the jump reappears at full size.

Evidence: power flat in ambient dimension at 0.74–0.76; clumpy-manifold honest FP 0.49 $\rightarrow$ 0.00; trend-hidden gaps unmasked 0.14 $\rightarrow$ 0.90. [\[VALIDATED\]](#)

Structural cost: 25% breakdown by construction, so $\pi \approx 0.5$ is out of reach—and at exactly 0.5, “majority branch” stops being a defined concept. [\[LIMIT\]](#)

9 Stage 3 — is this mixture attributable to a boundary?

Two tests in parallel at each rung, with a guard.

$$H_0 : r \sim N(\mu, \sigma^2) \quad \text{versus} \quad H_1 : r \sim \pi N(\mu_1, \sigma^2) + (1 - \pi) N(\mu_2, \sigma^2)$$

with σ *shared* between components.

The dip tests only “the density has one hump.” It assumes almost nothing and stays valid under curvature. Its floor is hard, and worth understanding rather than lamenting: an equal-variance two-component Gaussian mixture is *literally unimodal* until separation exceeds about 2 standard deviations. Below that, the dip’s measured power of 0.000 is not weakness—the alternative does not exist under its null.

The equal-variance LRT buys the sub-bimodal regime: 0.42–0.54 at 1.5 sd, 1.00 at 2.5 sd. The shared σ matters technically—an unrestricted normal-mixture LRT diverges—and statistically, since a narrower alternative has more power. It is close to efficient rather than merely better: the best single Gaussian matches the mixture’s first two moments exactly, so all signal lives in skewness and excess kurtosis, and power predicted from moment structure (0.55/0.30 at $\pi = 0.25/0.50$) matched observation (0.54/0.32).

Calibration must run through the whole pipeline. Resample inlier residuals, regenerate responses, rerun filter $\rightarrow$ trim $\rightarrow$ refit $\rightarrow$ residualise $\rightarrow$ both tests, $B = 300$. Trimming is a nonlinear, data-dependent operator, so anything calibrated against untrimmed theory is miscalibrated. In particular, **do not reuse a threshold derived from clean Gaussian draws**: LTS truncates the tails before the LRT sees them, which deflates the null. Costs zero model queries.

The separability guard—and this is where the estimator is born. A real gate is not merely a mixture in response space. It is a mixture whose components are **linearly separable in z** , because the boundary is locally a hyperplane; curvature artifacts produce components interleaved in space. So fit a linear classifier on (probe coordinates, responsibilities) and check held-out balanced accuracy: near-perfect for a gate, near chance for an artifact.

This closes the LRT’s one hole. The dip’s null is broad and survives curvature; the LRT’s null is “one Gaussian,” and curvature gives skewed-but-unimodal residuals that a mixture fits better. A power floor is traded for a robustness hole, and the guard is what covers it. The exposure is real: in a contaminated tabular setting the GLRT flagged honest models at 0.85 against the dip’s 0.52, so on clumpy data **the dip stays primary and every LRT-only flag is conditioned on this guard**.

And the guard’s decision boundary, lifted through the frame U , is the estimate of the gate’s normal. Test and recovery are the same computation. This is the hinge of the whole note.

PART III

The estimator

10 Stage 4 — what one anchor knows

Five outputs per anchor per rung. Three exist; two are new.

Gap. $\hat{\Delta} = |\hat{\mu}_1 - \hat{\mu}_2|$ from the mixture fit. Recovered 0.291 against a planted 0.300, honest-model baseline 0.014. [VALIDATED]

Composition. $\hat{\pi}$ from the same fit, oriented query-only: the anchor’s own response says which component it sits in, and the crossing mass is the other one. [VALIDATED]

Distance. [NEW] For a unit normal and probes $z \sim N(0, \sigma_s^2 I)$, the signed projection is $N(0, \sigma_s^2)$; crossing means the projection exceeds d . Hence

$$\hat{d}_s = -\sigma_s \Phi^{-1}(\hat{\pi}_s)$$

the crossing law read backwards. The law itself is [VALIDATED]—verified to three decimals in both its fixed-per-coordinate and fixed-total forms. Using it as an *estimator* is new, and as of 13 August it is measured:

1-D known-regimes setting. $\Delta = 0.30$, ladder `geomspace(0.02, 0.2, 3)`, $m = 1000$ per rung, with two query-only guards: the dip fires under Bonferroni, and $\hat{\pi} \in [0.05, 0.50]$. $\text{corr}(\hat{d}, d_{\text{true}}) = 0.9958$ ($n = 217$) · median relative error -0.92% · median absolute error 0.0031

Tracks correctly from $d = 0.012$ to 0.251, and is flat across a $10\times$ noise sweep: median absolute error 0.0029/0.0031/0.0027 at $\tau = 0.005/0.02/0.05$.

Two cautions carried by that result. Relative error degrades near the boundary—3.3% beyond $d = 0.1$, 5.0% at 0.05–0.1, 26% below 0.01—where the *absolute* error is still ~ 0.003 and only the denominator is small. Report absolute error near the boundary.

And **the estimate is conditional on detection**. Without the dip guard, anchors beyond the ladder top return confident nonsense (true 0.277 against estimated 0.141, with enormous spread), because $\hat{\pi}$ there is not a crossing fraction at all. The guard is query-only, so this is legitimate—but the conditioning belongs in the claim.

Direction. [VALIDATED] The separability classifier’s boundary, lifted to ambient space:

$$n_a = \frac{U\nu}{\|U\nu\|} \quad (\text{oriented: higher-mean component on the } + \text{ side}), \quad c_a = n_a^\top x_a + \frac{b}{\|\nu\|}.$$

Measured 13 August at intrinsic $d = 2$, ambient $D = 20$, frame supplied, 200 anchors per cell over a $\Delta/\tau \times \pi$ grid. The concern this had to answer is that responsibilities are *estimated*, not true branch labels, so the chain misassignment $\rightarrow$ classifier label noise $\rightarrow$ bias in $\hat{n}$ could dominate everything else near weak separation. It does not:

Misassignment costs variance, not bias. Mean signed rotation per cell: the oracle arm (true branch labels) lies in $[-0.35^\circ, +0.27^\circ]$; the estimated-responsibility arm lies in $[-6.6^\circ, +5.9^\circ]$

and is **not significant at 95% in any of the 20 cells**.

This is the symmetry argument holding: the responsibilities are a function of the one-dimensional residual, the residual is a function of $z^\top n$, and so the class-conditional means of z remain on n however badly the labels are corrupted. Corrupted labels *attenuate* the separation; they do not rotate it.

Because bias is what survives pooling and variance is not, this is the outcome that matters. Pooled orientation error falls as $1/\sqrt{N}$: at $\pi = 0.05$ it goes $6.8^\circ \rightarrow 0.9^\circ$ between $N = 1$ and $N = 100$ at $\Delta/\tau = 2.5$, and $20.4^\circ \rightarrow 1.8^\circ$ at $\Delta/\tau = 1.5$ — which is below the dip’s floor entirely.

Fit the discriminant on all probes, not on the trimmed inliers. [NEW] Median orientation error in the informative cells is 5.8° using every probe against 21.9° using the LTS inlier set. The trim removes points that are predominantly the minority branch — which is exactly the branch carrying the crossing signal. So: trim to obtain clean residuals for the *mixture*, then discard the mask for the *discriminant*. Soft responsibilities also beat hard $\gamma > 0.5$ labels (5.8° against 7.6°); keep the uncertainty.

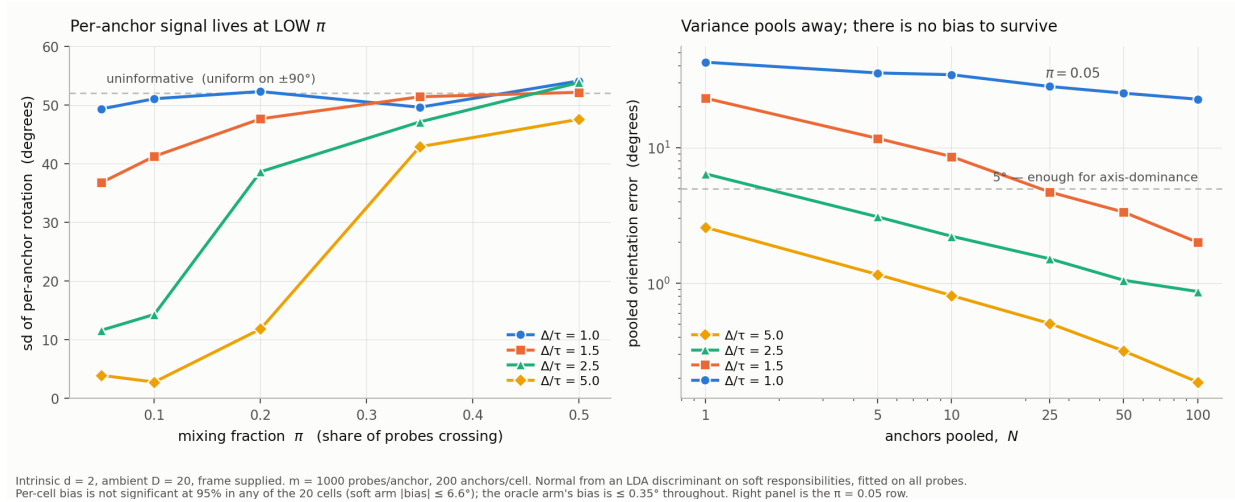


Figure 3: Left: per-anchor information against the mixing fraction. Signal lives at *low* π ; 52° is what uniformly random directions give. Right: pooled orientation error against the number of anchors, at $\pi = 0.05$.

Where the useful anchors are, and it is not where one would guess. [NEW] Per-anchor orientation error is **monotone increasing in π** , not U-shaped: at $\Delta/\tau = 5$ the per-anchor standard deviation runs 3.9° at $\pi = 0.05$ to 47.5° at $\pi = 0.50$. Fewer crossers, but they sit much further out along the normal, and the separation of class-conditional means along n *ris*es as π falls:

π	0.05	0.10	0.20	0.35	0.50
mean separation along n	2.17σ	1.95σ	1.75σ	1.63σ	1.60σ

At $\pi \rightarrow 0.5$ there is additionally no majority branch for LTS to commit to, so the plane lands between branches and corrupts the residuals the mixture sees. The useful anchors sit at $d \approx$

1.3–1.6 σ . This agrees with the distance estimator, which is also most accurate at larger d/σ , so a single rung serves both.

Weight. w_a is the LRT statistic, **not thresholded**. At per-anchor power near 0.5, thresholding before pooling discards half the true signal.

Every anchor emits $(n_a, c_a, \hat{d}_a, \hat{\Delta}_a, w_a)$ —**including anchors that did not fire**.

11 Stage 5 — three consistency checks, all free

The $S = 3$ ladder supports three checks on queries already spent. They test different things, and none is tuned.

(a) How the response scales with radius

Regress $\log \hat{\Delta}$ on $\log r$ across rungs; report the exponent α .

Structure	$\hat{\Delta}$ scaling	ratio at $r/2$	ratio at $r/4$
step (top-1 gate)	r^0	1.00	1.00
kink (top-2 softmax)	r^1	0.50	0.25
curvature (confound)	r^2	0.25	0.06

Observed: gates 1.01; resonant curvature 0.04 at $r/4$ against r^2 's predicted 0.0625. The log-log regression is preferable to a single-ratio threshold—free, more robust, and three-way rather than binary.

(b) Whether the geometry is self-consistent [NEW]

For a planar boundary at fixed distance, $\hat{d}_s$ should not depend on the rung. A gate returns the same $\hat{d}$ at every scale; a smooth confound's $\hat{\pi}_s$ is a mixture fit to something that is not a crossing, so its implied distance wanders. Measured 13 August:

Model	median CV of $\hat{d}$	fire rate
gated	0.098 (0.074 at $\tau = 0.02$)	0.678
GP null, $\ell = 0.05$, $A = 0.10$	0.557	0.397
GP null, $\ell = 0.05$, $A = 0.20$	1.042	0.114
kink control	1.241	0.023
honest	—	0.000
GP null, $\ell = 0.30$ (above ladder top)	—	0.000

At $CV < 0.15$: **56% of gates retained, 93% of confounders rejected**. As a standalone filter that is expensive; its value is as one of three orthogonal checks, and the operating point should be set after the full pipeline exists—trimming will sharpen $\hat{\pi}$ and will move it.

(c) Whether the two routes to the offset agree [NEW]

The offset is available twice: from the classifier's own intercept $b/\|v\|$, and from $-\sigma\Phi^{-1}(\hat{\pi})$ signed by $\hat{n}$. Both are functionals of the same responsibilities, so under a correctly specified

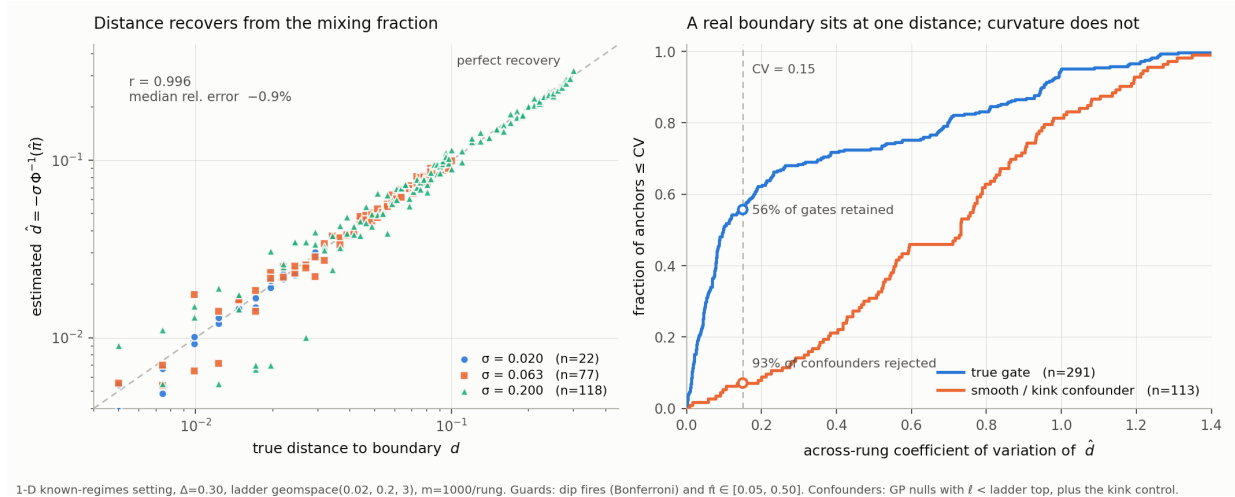


Figure 4: Left: the distance estimator against ground truth, by probe scale. Right: across-rung constancy of $\hat{d}$ separates a true gate from smooth and kink confounders.

model they agree by construction—but they *diverge differently* under misspecification, the way method-of-moments and maximum likelihood do. A large gap says the boundary is not planar inside the ball, or the mixture is spurious. This is a specification check, not two independent samples, and should be described as such.

Failing any of the three routes to `A5_structure_unattributed`—structure present, not attributable to a boundary. A third outcome: not a flag, and not a clean bill.

12 Stage 6 — pooling

[SPECIFIED] A boundary is a shared object: one hyperplane passes through many probe balls. Per-anchor power of 0.5 barely matters if evidence for a *common* boundary aggregates over 200 anchors—which means **per-anchor power is the wrong figure of merit**. The right one is the number of anchors needed to recover the boundary.

Do not threshold before pooling. All anchors contribute, weighted by w_a .

Estimator	Uses geometry
A Fisher/Stouffer combination of per-anchor p -values	no — baseline
B threshold on w_a , then cluster (n, c)	partly
C w -weighted mode-seeking (mean-shift / Hough) over all anchors	fully

Predicted ranking $C > B > A$ at every Δ , with the largest gap at $\Delta = 1.5$ sd, where half the true detections sit below any per-anchor threshold and B throws them away.

A no-information guard is required. [NEW] Outside the operating region — $\Delta/\tau = 1.0$ at any π , or $\pi \gtrsim 0.35$ — the per-anchor rotation has standard deviation $\approx 52^\circ$, which is exactly what uniformly random directions give. Pooling *axial* estimates that carry no information does not converge to the truth and does not converge to nothing: the mean direction of N random axes is itself uniformly random, so the pooled estimate lands on an arbitrary direction with an interval

that tightens around it. **Pooling can therefore be confidently wrong in a way that per-anchor abstention never is.** Stage 6 needs an explicit check that the anchor set carries information before it reports — the permutation null is the natural instrument, since uninformative input should be indistinguishable from permuted input.

Null. Permute the residual-to-point assignment within each anchor and rerun the pipeline. This randomises normals while preserving marginal residual distributions; ≥ 500 permutations.

Metrics. $N \mapsto$ orientation error, $N \mapsto |\hat{c} - c^*|$, and N_{95} . The first two are the estimation curves; N_{95} is the detection curve. Under this note’s framing, the first two are the headline.

One warning, stated as a prediction to register rather than a result. The naive argument for pooling—real boundaries give consistent normals, artifacts give random ones— **fails for the confound that matters.** A resonant surface $0.5 \sin(2\pi u_1 / \ell)$ has a preferred direction, so every anchor on it produces a normal aligned with $\pm u_1$. Those pool *coherently* into a confident false boundary.

The offsets should not agree, though. A gate places every c_a at the same value; a sine places them at its steep parts, spaced by ℓ :

gate $\longrightarrow$ a single mode in c , resonant curvature $\longrightarrow$ a comb of modes spaced by ℓ .

Register this before running. If it fails, smooth rejection returns to Stage 5 and the per-anchor decision rules are back on the critical path.

13 Stage 7 — stating the rule

```
axis-dominance:  is one |n_hat_i| dominant?
                  calibrate against the permutation
                  distribution of max_j |n_hat_j|

if yes:  coordinate i, threshold t_hat = c_hat / n_hat_i,
         CI from the bootstrap
if no:  report the hyperplane (n_hat, c_hat),
         no single-coordinate claim
```

PART IV

What must still be true

14 The obligation the reframe creates

A detector is validated by power and false-positive rate. **An estimator is validated by bias, variance, interval coverage, and query complexity**—and three of those four have never been measured here.

Coverage matters most and has been touched least. The ± 0.07 in the target sentence is doing enormous work: it is what turns a point estimate into a scientific claim. If the bootstrap interval on $\hat{f}$ does not cover at its nominal rate, that sentence is false in a way a reviewer will find immediately.

Calibrated coverage of the interval on $(\hat{n}, \hat{c}, \hat{t})$ is a first-class deliverable, not a finishing touch. Under the detector framing, nothing required it.

15 The critical path

1. **Validate $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$.** Done 13 August (§10).
2. **Validate the separability-derived normal.** Done 13 August (§10): no detectable bias in any cell, and pooled error below 3° at $N = 100$ down to $\Delta/\tau = 1.5$. Intrinsic $d = 2$ only — the sweep over $d \in \{4, 8\}$ is a parameter change in the same harness and remains to be run.
3. **Experiment P** $\rightarrow N \mapsto$ orientation error, $N \mapsto |\hat{c} - c^*|$, N_{95} . *Revise the distance axis:* the design specified anchor distance / $r_{hi} \in \{0.2, 0.5, 0.9, 1.5\}$, whose small values place anchors close to the boundary — which §10 shows is exactly where orientation is worst. Sample the low- π region instead.
4. **Axis-aligned threshold recovery with a calibrated interval.**

On deferring Stage 0. The order above proves geometry recovery first under *known-valid* local probes—a flat manifold of known dimension, where the frame and radius can be supplied rather than estimated—and adds automatic geometry selection afterwards. This is deliberate: the contribution is that $(\hat{n}, \hat{c})$ is recoverable at all, and establishing that under a hand-supplied probe geometry isolates it from any question about how the geometry was chosen. Stage 0 is then the extension that *removes* the assumption, converting “given a valid local probe geometry” into a condition the method checks for itself and declines when it fails. Deferred, not dropped: §6 remains the specification, and the results in steps 2–4 must state the conditioning explicitly. The one part of Stage 0/1 kept regardless is the **density filter**, because it scopes the claim rather than improving it.

Where the error budget lives — answered. *The prediction below was made before step 2 and is retained with its outcome.* Responsibilities are *estimated*, not true branch labels. The chain runs: mixture misassignment $\rightarrow$ classifier label noise $\rightarrow$ bias in $\hat{n}$. Near weak separation the responsibilities go diffuse ($\gamma \approx 0.5$ almost everywhere) and the discriminant direction destabilises; this may dominate finite-sample error entirely in the regime that matters. The clean way to isolate it is to **fit the classifier twice**—once on estimated responsibilities, once on true labels—and difference the orientation errors, as a function of Δ/τ and π , not only N . If that gap is large at $\Delta = 1.5$ sd, pooling must weight by *assignment confidence* rather than by the LRT statistic alone.

Outcome: the gap is entirely variance. Bias is not significant in any cell, so weighting by the LRT statistic is not disqualified, and the binding constraint turns out to be anchor *placement* ($\pi \leq 0.10$) rather than the weighting rule.

16 Scope: this is a locally single-boundary method

The procedure identifies **at most one boundary per probe ball**. Its precondition is $r_{hi} < L$, where L is the routing cell scale—and that precondition **is not checkable from the covariates alone**, because routing density lives in the response while covariate geometry is unchanged. [LIMIT]

When it is violated—many experts, small cells, a ball spanning several—the many small-separation components read as heteroscedastic noise, and **Stage 0 is blind. No abstention fires**. This is the only failure mode in the method that returns a confident wrong answer rather than declining to answer.

Two things follow. It explains why $K = 2$ globally is the right scope rather than an arbitrary simplification: the local procedure never sees more than one boundary regardless, so global K must be *assembled* from local evidence, never fitted as a large mixture. And it suggests a response-side detector that does not yet exist—if a ball spans several cells, $\hat{\Delta}$ may shrink with r rather than staying flat, which the existing ladder would see. Worth testing in the step-2 harness; not claimed. [OPEN]

17 Everything else that limits the claim

- **Parallel penalty.** An offset aligned with strong within-branch variation is invisible at every radius, in every dimension, under every projection. Proven, not conjectured. [LIMIT]
- **Reach is not validity.** The plateau says where assumptions hold, not where the gate lives. [LIMIT]
- $\pi \approx 0.5$. LTS breakdown is structural. The LRT fixes *testing* (0.32, graceful); recovery still breaks, since there is no majority branch to trim toward. [LIMIT]
- **Detection floor** ≈ 1.5 sd at $m \sim 10^3$. Moment z-scores scale like $\sqrt{n}$, so halving the detectable separation costs $4\times$ the queries.
- **Coverage.** Randomly placed anchors contain a boundary about r_{hi}/L of the time—roughly 15–20%. Fine for detection; a real constraint on constructive recovery. Boundary-seeking placement does not exist and is a *deployment* problem: in simulation the boundary is known, so anchors are placed by design. [OPEN]
- **Validation scope.** One model class, one D , one τ_y , and no full-pipeline bootstrap yet. [OPEN]

18 Provenance summary

Component	Status
Crossing law; fixed-per-coordinate budget mandatory	[VALIDATED] to three decimals
Trimmed residualisation, dimension-independent power	[VALIDATED] 0.74–0.76 flat in D
OLS absorbs ~64% of the gap	closed form, matches measurement
Trimming restores validity on clumpy manifolds	[VALIDATED] 0.49 $\rightarrow$ 0.00
Dispersion cannot discriminate a gate from curvature	[LIMIT]
Dip floor ~ 2 sd; LRT 0.42–0.54 at 1.5 sd	measured; LRT uncalibrated through pipeline
Local $\hat{\Delta}$ recovery	[VALIDATED] 0.291 vs 0.300
Group-conditional estimand at the anchor	[VALIDATED] cosine 1.000
Distance estimator $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$	[VALIDATED] $r = 0.996$, median rel. error -0.9%
Distance invariance across rungs	[VALIDATED] CV 0.098 vs 0.56–1.24
Round-consistency plateau; rank-decade W ; A1/A1n split	[NEW], unimplemented
Stage 0 entire (GRIDE, plateau, frame)	[SPECIFIED] — no implementation exists
Boundary normal from the separability classifier	[VALIDATED] no bias in 20/20 cells
Pooled orientation error $< 3^\circ$ at $N = 100$, $\Delta/\tau \geq 1.5$	[VALIDATED] (at $d = 2$)
Discriminant on all probes, not LTS inliers	[NEW] 5.8° vs 21.9°
Orientation error monotone increasing in π	[VALIDATED] — prediction of a U-shape falsified
No-information guard for pooling	[NEW] — required, not yet designed
Intrinsic-dimension scaling of orientation error	[OPEN] — $d = 2$ only
Offset-agreement check	[NEW]
Pooling, permutation null, N_{95}	[SPECIFIED] — Experiment P
Comb-versus-single-mode discriminator	[NEW] — register before running
Axis-dominance and threshold recovery	[NEW]
Interval coverage on $(\hat{n}, \hat{c}, \hat{t})$	never measured — §14